

PROJECTS

- **Leviathan**

Searchless Transformer Chess Engine · PyTorch, FastAPI, Next.js, TypeScript, Tailwind CSS

HuggingFace Spaces
Jan 2025 – Dec 2025

- Designed and trained 20M-parameter transformer with custom attention for real-time inference on edge hardware (RTX 4060, 8GB VRAM), optimizing architecture and training for resource-constrained deployment.
 - Reduced KV-cache memory footprint by 8× through Multi-Head Latent Attention implementation, enabling batch inference and 19ms per-move latency in real-time inference settings.
 - Designed custom tokenizer encoding board states and UCI notation into 4,247-token vocabulary for single-token move prediction.
 - Trained on 3M+ chess positions using gradient checkpointing and mixed-precision (FP16) to maximize data throughput under memory constraints.
- **Multi-Head Latent Attention (MLA)**

Research Implementation · PyTorch

GitHub ● PyPI
Jun 2025 – Aug 2025

- Translated Multi-Head Latent Attention from DeepSeek-V2 research into modular PyTorch implementation, designing clean abstractions for KV compression and low-rank projection components.
 - Packaged implementation as production-ready PyPI library with type hints, documentation, and integration examples.

EXPERIENCE

- **NetApp Inc.**

Software Engineer in Test

Wichita, KS
Aug 2024 – Present

- Built automated analysis pipeline processing storage telemetry across distributed RAID systems (FC, SAS, NVMe/RoCE), handling terabytes of performance data.
 - Designed Python automation framework reducing manual configuration tasks by 30% across Linux, Windows, and VMware infrastructure.
 - Implemented Ansible-based deployment orchestration for 300+ system configurations, streamlining infrastructure provisioning workflows.
 - Developed interactive visualization dashboard for system performance metrics using Python, analyzing 1M+ database entries for engineering insights.

EDUCATION

- **Wichita State University**

Accelerated Master of Science in Computer Science

Wichita, KS
Jan 2026 – Dec 2027 (Expected)
- **Wichita State University**

Bachelor of Science in Computer Science; GPA: 3.66/4.0, Magna Cum Laude

Wichita, KS
Jan 2022 – Dec 2025

- Dean’s List:** Spring 2022 – Fall 2025
 - Relevant Coursework:** Machine Learning, Artificial Intelligence, Fundamentals of AI Agents, Data Science

SKILLS

- **ML/DL:** PyTorch, Transformers, Hugging Face, Model Training, Fine-tuning, Attention Mechanisms, Gradient Optimization
- **Data & Computation:** NumPy, Pandas, Scikit-learn, Matplotlib, Mixed-Precision Training
- **Languages:** Python, C++, C, SQL, JavaScript, TypeScript
- **Tools & Systems:** Git, Docker, Linux, Ansible, Jupyter, FastAPI